

Emanuel Munoz

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EDUCATION

Carnegie Mellon University (CMU), Robotics Institute

Master of Science in Robotics | GPA 4.05

- *Relevant coursework: Embodied Safe AI, Computer Vision, Robot Learning, Multi-agent planning*

Pittsburgh, PA

August 2026

University of Engineering and Technology, UTEC

Bachelor of Science in Electrical Engineering | GPA: 4.0

- *Thesis: Active Robotic Exploration via PPO Reinforcement Learning and Self-Supervised Learning*

Lima, Peru

December 2021

SKILLS

Programming & ML: Python, C/C++, PyTorch, Tensorflow, ROS/ROS2, OpenCV.

Deployment & Tools: AWS, Docker, GCP, Git, Linux, PostgreSQL, CUDA, MuJoCo, Gazebo, Jetson Orin.

EXPERIENCE

Carnegie Mellon University (CMU), Robotics Institute

Graduate Research Assistant

Pittsburgh, PA

August 2024 – August 2026

- Developed a real-time pipeline using Gaussian Processes to model uncertainty and mitigate visual sensing impairment risks, enabling navigation in smoke-dense environments.
- Designed and implemented an online autoregressive probabilistic predictor using Neural Operators for spatiotemporal behavior forecasting of highly dynamic smoke, enabling hazard avoidance.
- Engineered a hierarchical risk-aware path planning framework for multi-robot navigation in hazardous areas, reducing dense smoke exposure by up to ~85% compared to baselines ([Accepted, IROS 2026](#)).

Noteworthy AI

Graduate Machine Learning Engineer Intern

Connecticut, CT | Remote

May 2025 – August 2025

- Prototyped Out-of-Distribution (OOD) detectors for ViT (Vision Transformer) models to automatically identify unseen image samples, achieving up to ~70% recall for automated incremental training.
- Fine-tuned ViT models using long-tailed datasets, reaching ~90% F1-score in object grid defect classification.

Noteworthy AI

Machine Learning Engineer

Connecticut, CT | Remote

March 2022 – August 2024

- Developed and deployed a real-time edge pipeline using C++ and ROS2 on Jetson Orin to capture georeferenced telemetry and image data from utility poles, enabling automated infrastructure inspection via fleet vehicles.
- Engineered a 3D visual tracking pipeline integrating a YOLO detector, stereo depth imaging, and IMU/GNSS sensor data for spatial state localization of target assets, used for physical-to-digital asset correlation.
- Designed a low-latency triggering framework using model predictive tracking to intercept target assets from moving vehicles, enabling automated, high-resolution industrial camera capture.
- Built cloud-based ML pipelines on AWS using PyTorch for training, inference, evaluation, and benchmarking of production-level ViT models for asset detection and defect classification.

Carnegie Mellon University (CMU), Robotics Institute, RISS Program

Research Scholar

Pittsburgh, PA | Remote

May 2021 – July 2022

- Designed a safety-critical motion planning framework for self-driving vehicles, integrating barrier functions, optimal control, and adaptive learning to mitigate dynamic uncertainties.
- Validated framework robustness within the CARLA simulation environment, demonstrating zero safety violations under dynamic uncertainty; research published at [IROS 2022](#).

University of Engineering and Technology, UTEC - CONCYTEC

Research Assistant

Lima, Peru

March 2020 – April 2021

- Engineered a feature-based multi-stereo camera registration pipeline to reconstruct a surgical environment for teleoperation tasks, delivering real-time 3D point cloud representations.
- Developed a hybrid learning controller to map torque signals onto position-based manipulators, enabling full torque controllability and mitigated tracking errors ([CDC 2020](#)).

SELECTED PROJECTS

Motion-Token MAPF: Multi-Agent Coordination in Mixed Traffic | [\[GitHub\]](#) | [\[Slides\]](#)

CMU, 2026

- Developed a multi-agent framework using an autoregressive token-based motion predictor with Priority-Based Search to coordinate safe autonomous vehicles amidst human drivers, reducing collision rates and travel times.

PINN-Neural Process: Few-Shot Adaptation Framework for Physics-Informed ML | [\[GitHub\]](#)

Independent, 2026

- Developed a physics-informed meta-learning framework with conditional, variational, and flow matching-based Neural Process variants to eliminate the retraining bottleneck of standard PINNs, enabling few-shot adaptation.

DreamerV2 in ViZDoom: Model-Based Reinforcement Learning | [\[GitHub\]](#)

CMU, 2025

- Implemented DreamerV2 world model architecture with a self-supervised DINO backbone to evaluate latent representations from 3D first-person inputs within the ViZDoom environment for benchmarking.